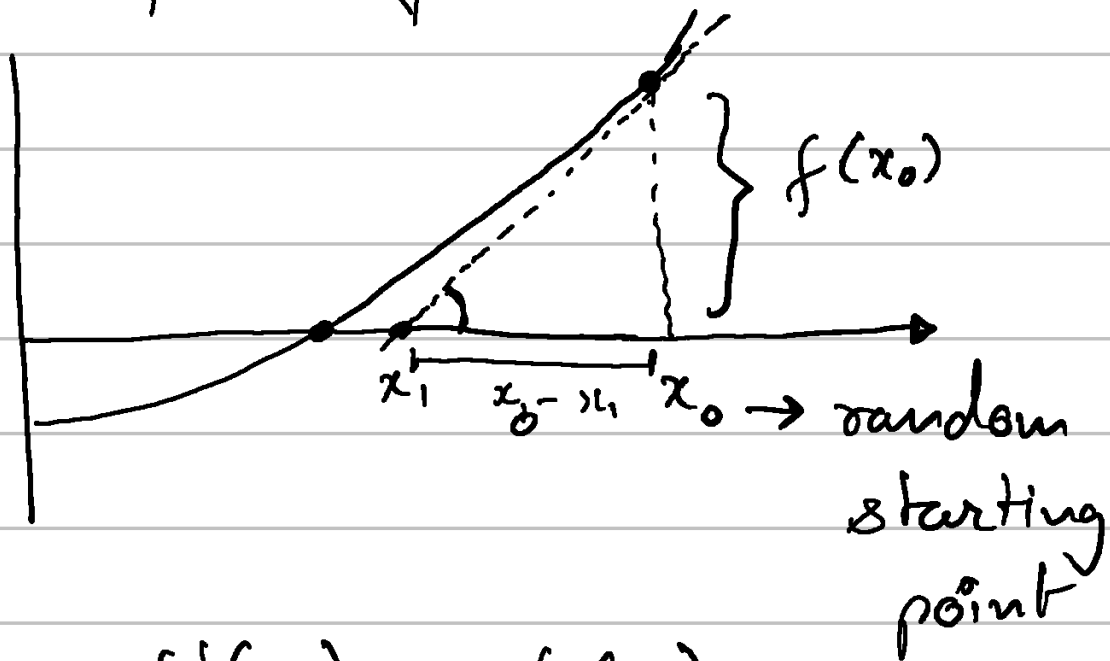


Lesson 6 - Newton's Method

⇒ Newton's method is used to find zero of a function



$$\text{so, } f'(x_0) = \frac{f(x_0)}{x_0 - x_1}$$

$$\begin{aligned} f'(x_0)(x_0 - x_1) &= f(x_0) \\ \frac{x_0 \cdot f'(x_0) - f(x_0)}{f'(x_0)} &= x_1 \end{aligned}$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

⇒ Multiple iterations of this step yields points closer & closer to zero.

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_n = x_0 - \left[\frac{f(x_0)}{f'(x_0)} + \frac{f(x_1)}{f'(x_1)} \dots \frac{f(x_{n-1})}{f'(x_{n-1})} \right]$$

⇒ Now if we use Newton's method to find zeroes of derivative of a function, we can find the minima of that function.

eg $f(x) = e^x - \ln(x)$

$$\Rightarrow f'(x) = e^x - \frac{1}{x}$$

$$\Rightarrow f''(x) = e^x + \frac{1}{x^2}$$

$$\Rightarrow x_0 = 0.05$$

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)} = 0.097$$

$$x_2 = 0.183$$

* using a
calc here

$$x_3 = 0.32082$$

$$x_4 = 0.477$$

$$x_5 = 0.557$$

$$x_6 = \underline{0.567}$$

$$x_7 = \underline{0.567}$$

Second Derivative

$$\Rightarrow f''(x) \quad / \quad \frac{d^2 y}{dx^2}$$

→ Rate of change of "slope"

⇒ Second derivative tells us about the shape of a function at some point.

eg if $f''(x) < 0$



$f'(x)$ is decreasing.



$f(x)$ is approaching maxima



concave down

$$\begin{array}{lll} f''(x) > 0 & \rightarrow & \text{minima} \\ f''(x) < 0 & \rightarrow & \text{maxima} \end{array}$$

Hessian

Matrix of various second derivatives of multivariable function.

$$f(x, y)$$

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{yx}(x, y) & f_{yy}(x, y) \end{bmatrix}$$

Concavity & Hessians

⇒ In this case concavity is determined by positiveness of the Hessian matrix.

⇒ We can look at the eigenvalues of the diagonal entries to see how many positive values are present.

H

positive definite

negative definite
(all $\lambda < 0$)

Some $\lambda > 0$ & other $\lambda < 0$
(if $\lambda = 0$)

Shape

Concave up.
(minima)

concave down.
(maxima)

saddle point
(inconclusive when $\lambda = 0$)

Optimization using Hessian $\rightarrow$

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \end{bmatrix} - H^{-1}(x_k, y_k) \nabla f(x_k, y_k)$$

eg $f(x, y) = x^4 + 0.8y^4 + 4x^2 + 2y^2 - xy - 0.2x^2y$

$$f'_x(x, y) = 4x^3 + 8x - y - 0.4xy$$

$$f'_y(x, y) = 3.2y^3 + 4y - x - 0.2x^2$$

$$\bullet f''_{xy}(x, y) = -1 - 0.4x$$

$$\bullet f''_{xx}(x, y) = 12x^2 + 8 - 0.4y$$

$$\bullet f''_{yy}(x, y) = 9.6y^2 + 4$$

$$\cdot f''_{xy}(x, y) = -1 - 0.4x$$

$$f''_{xx}(x, y) = 12x^2 + 8 - 0.4y$$

$$f''_{yy}(x, y) = 9.6y^2 + 4$$

$$\nabla f = \begin{bmatrix} 4x^3 + 8x - y - 0.4xy \\ 3.2y^3 + 4y - x - 0.2x^2 \end{bmatrix}$$

$$H = \begin{bmatrix} 12x^2 + 8 - 0.4y & -1 - 0.4x \\ -1 - 0.4x & 9.6y^2 + 4 \end{bmatrix}$$

$$\text{Let } \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

$$X_1 = \begin{bmatrix} 2.58 \\ 2.62 \end{bmatrix} \rightarrow \rightarrow \text{need 8 steps}$$

$$\begin{bmatrix} x_8 \\ y_8 \end{bmatrix} = \begin{bmatrix} 4.15 \cdot 10^{-17} \\ -2.08 \cdot 10^{-17} \end{bmatrix}$$